

CSP650 FINAL YEAR PROJECT

Topic 3 - Writing Research Methodology

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(Source : (1) Essentials of Computing Sciences: Project Administration;
(2) Nur Atikah Sia & Suryani, CS)

Chapter 3 : Research Methodology

- Objective: To describe the overall process flow of the research (framework of the research, research structure)
 - To describe the techniques and procedure (steps) used in research; e.g. Steps in developing the algorithm, steps in pre-processing the dataset
 - To explain the data, materials, subjects, tools and equipment used in research
 - To explain how the experiments are designed
- Generally to describe:
 - *How the project/research is carried out?*

Guidelines in writing Research Methodology

- **Refine** the proposal RM
- Begin with the overview of RM – all phases in one figure
 - Research framework/structure (show the relationship between the objective, activities and deliverable)
- Explain every phase in detail

Content of Research Methodology- Sample A

3.1 Introduction	26
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3.8.2 Software Requirements	57
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Content of Research Methodology- Sample B

3.1 Project Framework

3.2 Theoretical Study

3.3 Empirical Study

3.3.1 Knowledge Elicitation

3.3.2 Data Collection

3.3.3 Data Analysis

3.4 Expert System Architecture Construction

3.4.1 Knowledge Design

3.4.2 System Design

3.5 System Development

3.6 Hardware and Software Specification

3.7 Research Plan

3.8 Summary

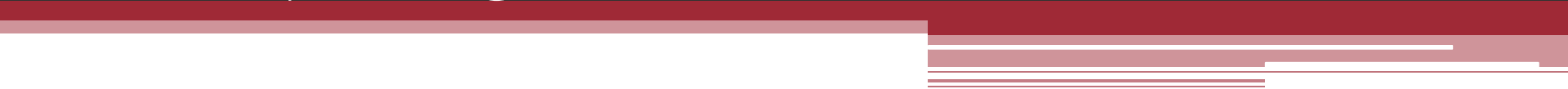
Content of Research Methodology- Sample C

- 3.1 Introduction
- 3.2 Theoretical Study
 - 3.2.1 Preliminary Study (produce project structure/framework)
 - 3.2.2 Knowledge Acquisition (interview results, gap in literature)
- 3.3 Empirical Study
 - 3.3.1 Data Collection
 - 3.3.2 Data Pre-processing
 - 3.3.3 Data Analysis
- 3.4 System Design
 - 3.4.1 Database Design
 - 3.4.2 Interface Design
 - 3.4.3 Architecture Design
- 3.5 System Development
 - 3.5.1 System Flow
 - 3.5.2 System Testing and Evaluation
 - 3.5.3 Hardware and Software requirements
- 3.6 Summary

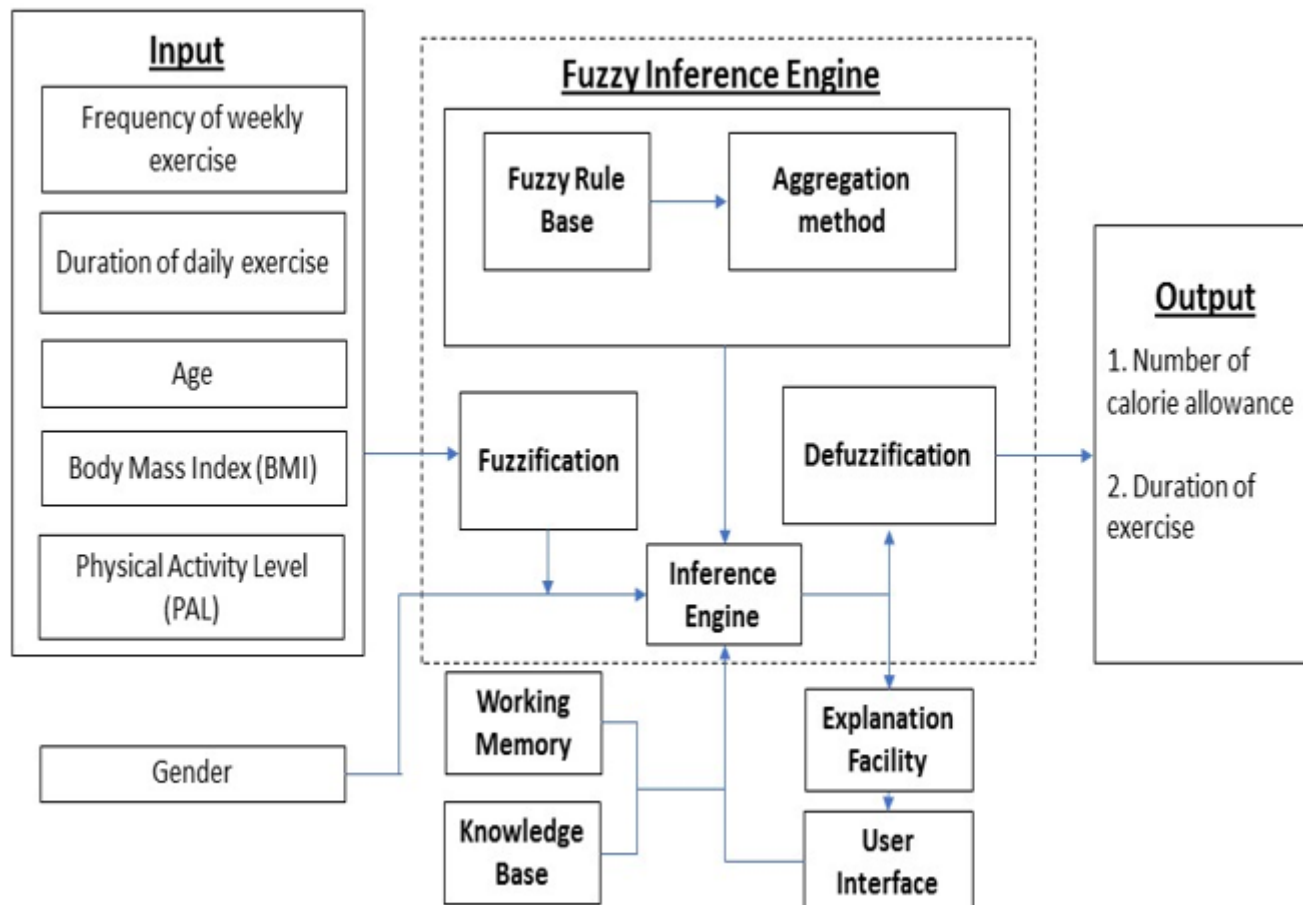
Chapter 3 - Tips

- Provide enough detail for replication of your work.
- Order procedures chronologically.
- Use past tense to describe what you did.
- Don't mix results with procedure.

EXAMPLES

A series of horizontal lines in red and white, some solid and some double-lined, extending across the bottom of the slide.

CALORIE CONTROL FOR THE HEALTHY SOCIETY USING FUZZY EXPERT APPROACH



System Architecture (Amirul, 2018)

VIRTUAL GROCERIES SHOPPING PATH SCHEDULER OPTIMIZER

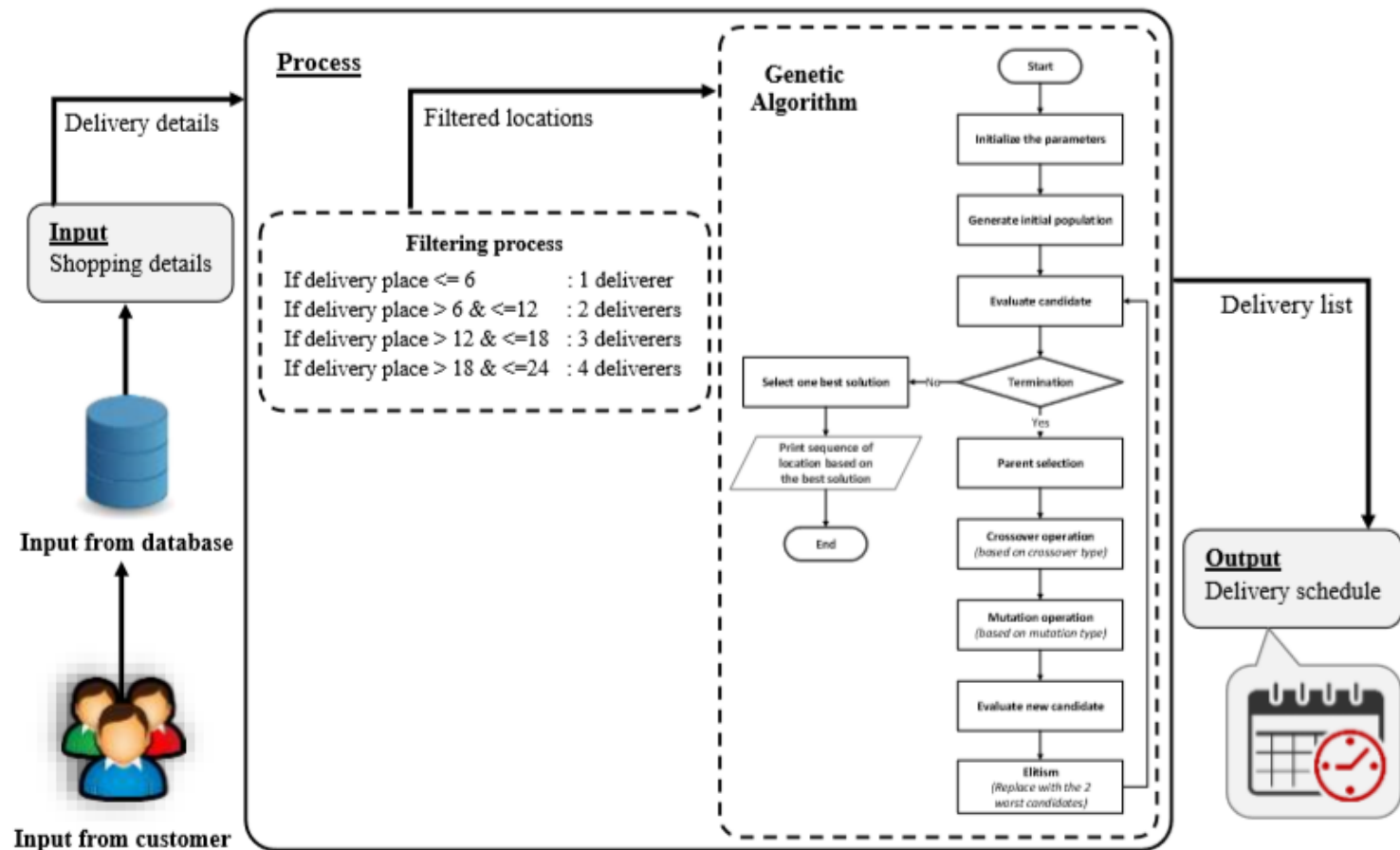


Figure 3.9: V-GROS System Architecture